

GNN for Molecules and SMILES

Visualization using OGBG-MolHIV

Project Report

Author: Revanth S

B.Tech Artificial Intelligence & Data Science

Amrita Vishwa Vidyapeetham

Abstract

This project demonstrates Graph Neural Networks (GNNs) for learning molecular graph representations using the OGBG-MolHIV dataset. A Graph Convolutional Network (GCN) is implemented using PyTorch Geometric and trained on molecular graphs. The project also visualizes molecular graphs and reconstructs molecular structures from SMILES strings using RDKit.

Objectives

- Understand graph representation of molecules.
- Train a Graph Convolutional Network.
- Visualize molecular graphs.
- Convert SMILES strings into molecular structures.
- Analyze training loss.

Introduction

Molecules can naturally be represented as graphs where atoms are nodes and chemical bonds are edges. Graph Neural Networks extend deep learning to graph-structured data, making them suitable for molecular learning tasks. This project provides a practical introduction to GNNs using the OGBG-MolHIV dataset.

Dataset

Dataset: OGBG-MolHIV

Nodes represent atoms, edges represent chemical bonds, and each graph corresponds to one molecule. The dataset is accessed through the Open Graph Benchmark library.

Methodology

1. Load the OGBG-MolHIV dataset.
2. Split into train, validation and test sets.
3. Build a two-layer GCN.
4. Train using the Adam optimizer.
5. Compute Binary Cross Entropy with Logits Loss.
6. Plot training loss.
7. Visualize a molecular graph.
8. Generate a molecular structure from a SMILES string using RDKit.

Model Architecture

- Two GCNConv layers
- ReLU activation
- Global Mean Pooling
- Fully Connected output layer
- BCEWithLogitsLoss
- Adam optimizer

Technologies Used

Python

PyTorch

PyTorch Geometric

Open Graph Benchmark (OGB)

RDKit

NetworkX

Matplotlib

Pandas

Results

The model successfully learns graph representations as indicated by decreasing training loss. The project generates a training loss plot, molecular graph visualization, and SMILES-based molecular structure visualization.

Applications

Drug Discovery

Computational Chemistry

Cheminformatics

Bioinformatics

Molecular Analysis

Future Scope

Evaluate on test data

Compute ROC-AUC

Compare GCN with GAT, GraphSAGE and GIN

Use additional molecular datasets

Improve visualization pipeline

Conclusion

The project demonstrates graph representation learning using Graph Neural Networks and introduces molecular graph processing and SMILES visualization using modern deep learning libraries.

References

1. Open Graph Benchmark
2. PyTorch Geometric Documentation
3. RDKit Documentation
4. Kipf & Welling (2017), Semi-Supervised Classification with Graph Convolutional Networks